Yelp Restaurant Performance & Customer Engagement Analysis

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1. Introduction

The restaurant industry is highly competitive, and understanding customer preferences is critical for business success. Online review platforms such as Yelp provide large volumes of customer-generated data that can help analyze restaurant performance and customer engagement patterns.

This project focuses on analyzing restaurant performance using Yelp data. By examining customer ratings, reviews, tips, and check-ins, the analysis aims to understand how customer engagement relates to restaurant popularity and success.

The project uses a combination of **Python, SQL, and Power BI** to transform raw data into meaningful insights. Python was used for data cleaning and preprocessing, SQL queries were used to aggregate data and generate summary metrics, and Power BI was used to create an interactive dashboard for visual analysis.

The final dashboard allows users to explore restaurant performance, geographic distribution, and customer engagement patterns.

2. Problem Statement

The restaurant industry relies heavily on customer feedback and engagement to evaluate business performance and customer satisfaction. Online platforms such as Yelp generate large volumes of customer interaction data, including ratings, reviews, tips, and check-ins. However, this raw data is often complex and difficult to interpret without proper analysis.

Restaurant owners, analysts, and business stakeholders need a clear understanding of how customer engagement and ratings influence restaurant success. Identifying patterns in customer behavior can help determine which restaurants perform well, which locations have higher customer activity, and how engagement metrics relate to restaurant popularity.

The main problem addressed in this project is to analyze restaurant performance and customer engagement using Yelp data in order to identify meaningful patterns and insights. Specifically, the project aims to understand how customer ratings, reviews, tips, and check-ins contribute to restaurant success and how these factors vary across different cities.

To address this problem, data analytics techniques were applied using Python for data preparation, SQL for data aggregation, and Power BI for interactive visualization. The resulting dashboard provides a comprehensive view of restaurant performance and customer engagement trends.

3. Project Objectives

The primary objective of this project is to analyze restaurant performance and customer engagement using Yelp data and to transform raw data into meaningful business insights through data analytics techniques.

The specific objectives of the project include:

1. **Analyze restaurant performance** by examining customer ratings and identifying top-performing restaurants.
2. **Evaluate customer engagement patterns** using key interaction metrics such as reviews, tips, and check-ins.
3. **Identify geographic trends** by analyzing restaurant distribution and performance across different cities.
4. **Examine the relationship between restaurant ratings and customer engagement** to understand how quality influences popularity.
5. **Develop an interactive Power BI dashboard** that allows users to explore restaurant performance and customer behavior through visual analytics.

4. Dataset Description

The dataset used in this project is derived from the Yelp dataset and includes information about restaurants and customer interactions.

The dataset contains information such as:

- Business ID
- Restaurant name
- City and state
- Average customer rating
- Review count
- Tip count
- Customer check-ins

These attributes allow the analysis of both **restaurant quality and customer engagement behavior**.

Key engagement indicators include:

- Number of customer reviews
- Tips left by customers
- Check-in activity

These metrics provide insights into how actively customers interact with restaurants.

5. Data Cleaning and Preparation (Python)

Python was used to prepare and clean the raw dataset before analysis. Data preparation ensures that the data is accurate, structured, and ready for analysis.

During the data preparation stage, the large Yelp dataset was filtered to include only restaurant-related businesses. This step was necessary because the original dataset contains multiple business categories such as shopping, services, and entertainment. By selecting only restaurant records, the analysis was able to focus specifically on restaurant performance and customer engagement patterns.

The following steps were performed during the preprocessing stage:

- Filtering the dataset to include only restaurant-related businesses
- Removing missing or inconsistent values
- Selecting relevant attributes such as ratings, reviews, and location information
- Aggregating customer engagement metrics
- Creating new calculated fields used in the analysis

Additional derived features were created to support analysis in the dashboard.

For example:

Rating Segmentation

Restaurants were categorized based on their ratings:

- High Rated (rating ≥ 4)
- Medium Rated (rating between 3 and 4)
- Low Rated (rating < 3)

This classification allows comparison of customer engagement across different restaurant quality levels.

6. Data Aggregation and Feature Engineering (SQL)

After cleaning the dataset, summary tables were created to support efficient analysis in Power BI, SQL queries were used to generate summary tables containing aggregated metrics.

These tables included:

- Restaurant-level metrics
- City-level metrics
- Customer engagement indicators

Key metrics calculated include:

Total Reviews

Total number of customer reviews received by each restaurant.

Total Tips

Number of tips provided by customers.

Total Check-ins

Number of customer visits recorded through check-ins.

Success Score

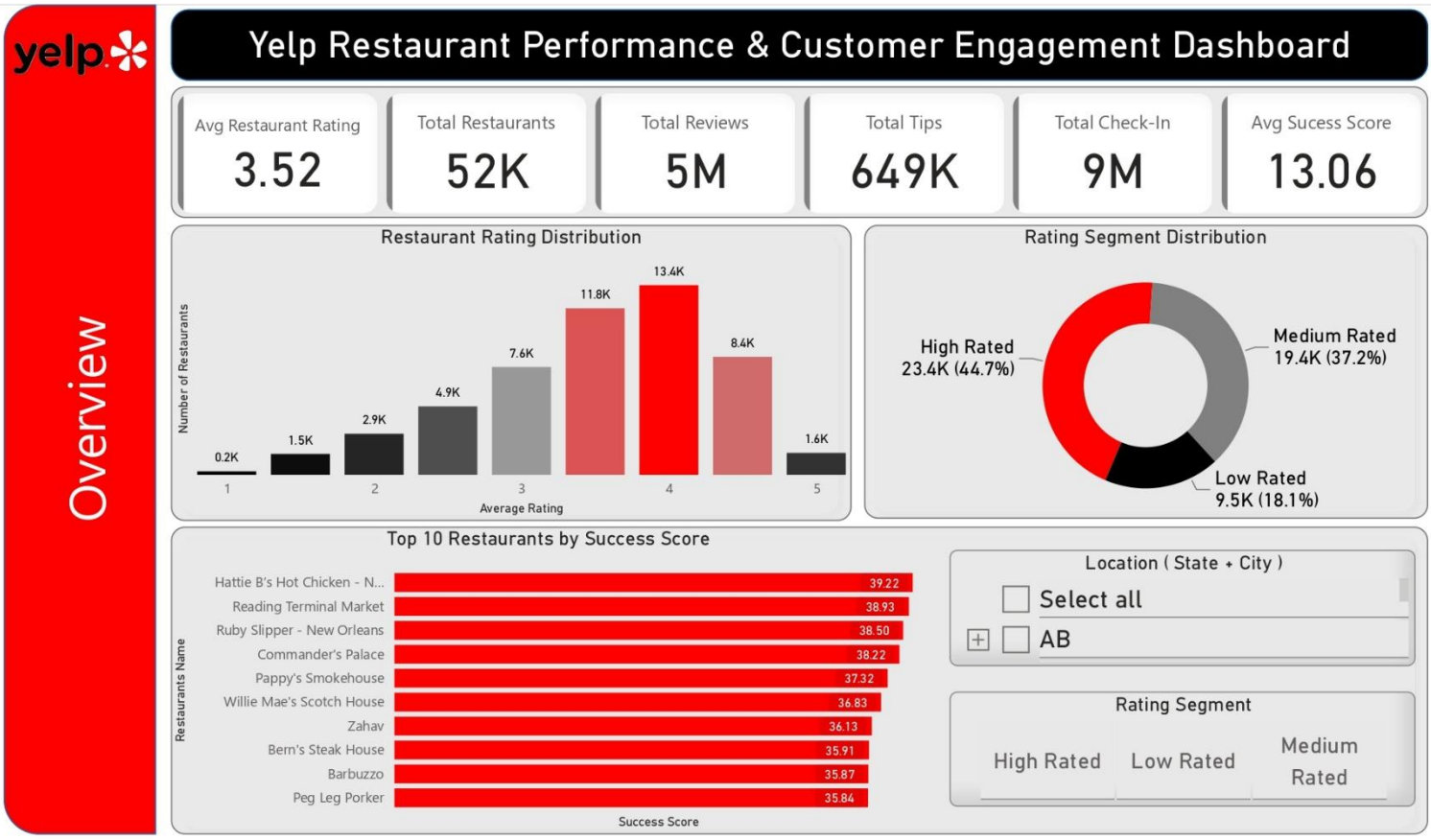
A metric designed to represent restaurant performance by combining customer ratings and engagement metrics.

The success score helps identify restaurants that perform well in both quality and popularity.

These aggregated datasets were then imported into Power BI to create the dashboard.

7. Power BI Dashboard Analysis

The final Power BI dashboard was designed to present insights about restaurant performance and customer engagement. The dashboard consists of **four analytical sections** that explore different aspects of the dataset.



7.1 Overview Dashboard

The overview page provides a summary of restaurant performance.

Key metrics include:

- Average restaurant rating: **3.52**
- Total restaurants: **52K**
- Total reviews: **5M**
- Total tips: **649K**
- Total check-ins: **9M**
- Average success score: **13.06**

The rating distribution chart shows that most restaurants fall within the **3 to 4 star rating range**, indicating moderate customer satisfaction across the dataset.

The dashboard also highlights the **top-performing restaurants based on success score**, which represent businesses that achieve both strong ratings and high customer engagement.

These metrics provide a quick overview of restaurant performance across the dataset.

7.2 Restaurant Performance Analysis

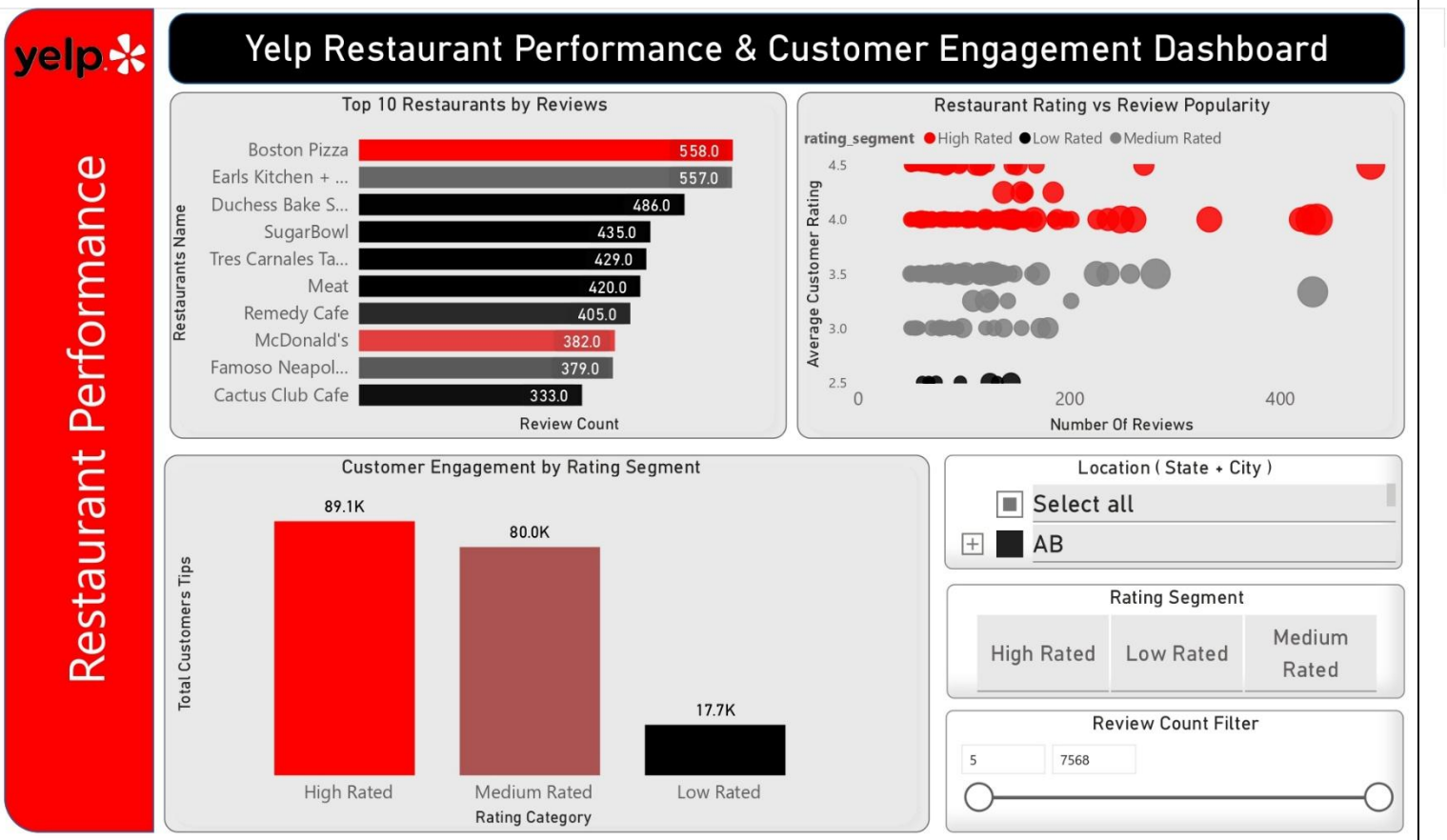
This section of the dashboard focuses on analyzing restaurant popularity and quality.

The analysis highlights the **top 10 restaurants based on review counts**, which indicates restaurants that receive the most customer attention.

A scatter plot comparing **customer ratings and review counts** provides insights into the relationship between quality and popularity.

The visualization shows that restaurants with higher ratings generally tend to attract more reviews, indicating that customer satisfaction can drive engagement and visibility.

Another visualization analyzes **customer engagement across rating segments**, showing that highly rated restaurants typically receive more reviews and tips compared to lower-rated establishments.



7.3 City Performance Analysis

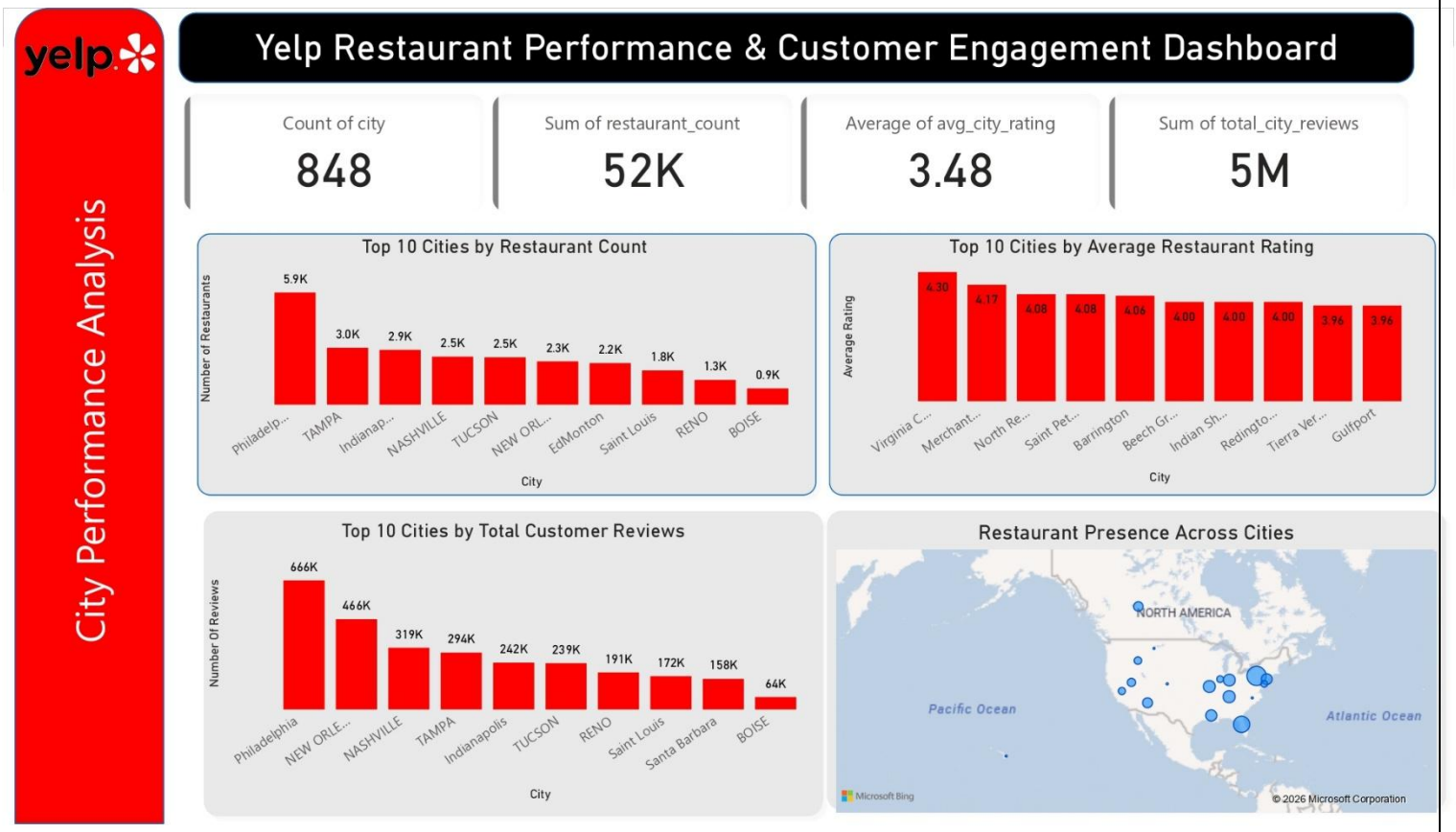
Geographic analysis was performed to understand restaurant distribution and customer activity across different cities.

The dashboard shows the **cities with the highest number of restaurants**, indicating areas with strong restaurant presence.

Cities such as **Philadelphia** appear as major restaurant hubs with a large number of businesses and high customer interaction levels.

The analysis also identifies **cities with the highest average restaurant ratings**, indicating regions where restaurants consistently receive positive customer feedback.

This geographic analysis helps highlight areas with high restaurant activity and customer engagement.



7.4 Customer Engagement Analysis

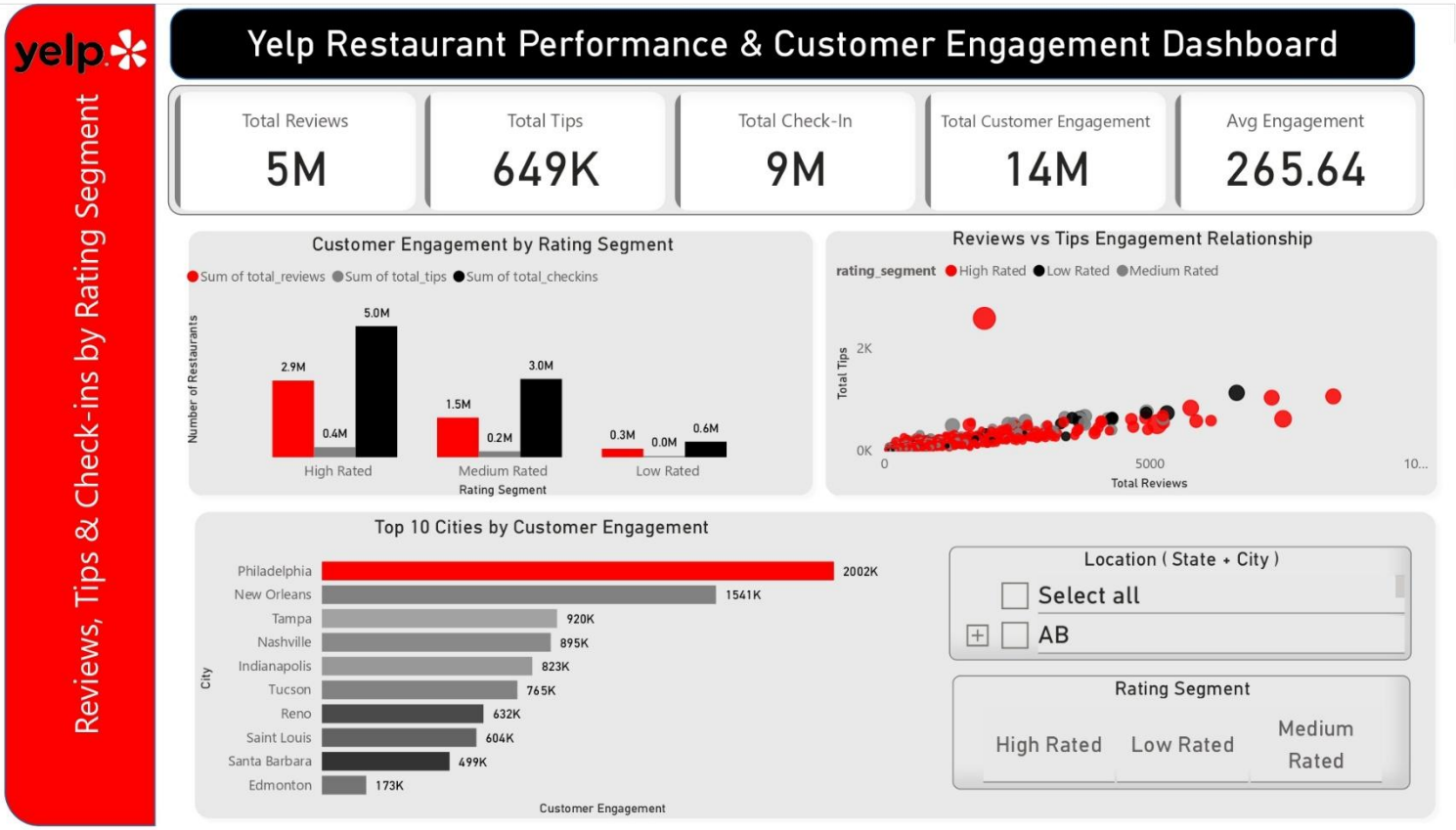
Customer engagement is measured through:

- Reviews
- Tips
- Check-ins

The dashboard indicates that:

- Total customer engagement reached approximately **14 million interactions**.
- High-rated restaurants generate the highest levels of engagement.
- Cities such as Philadelphia, New Orleans, and Tampa have the most customer interaction.

A scatter plot comparing **reviews and tips** shows that restaurants with more reviews generally receive more tips, indicating stronger customer engagement.



8. Key Insights

Several important insights were identified during the analysis:

1. Most restaurants fall within the **3 to 4 star rating range**, indicating moderate customer satisfaction.
2. Restaurants with higher ratings generally receive more reviews and tips from customers.
3. Cities such as **Philadelphia** show the highest concentration of restaurants and customer interactions.
4. Customer engagement metrics such as reviews and tips are strongly correlated.

Restaurants with strong customer engagement typically achieve higher success scores.

These insights demonstrate the importance of customer feedback and engagement in determining restaurant success.

9. Conclusion

This project demonstrates how data analytics tools can be used to analyze restaurant performance and customer engagement patterns.

By combining Python for data preparation, SQL for data aggregation, and Power BI for visualization, the project transformed raw Yelp data into meaningful business insights.

The interactive dashboard allows users to explore restaurant performance, identify top-performing restaurants, and understand geographic trends in the restaurant industry.

Customer engagement metrics such as reviews, tips, and check-ins play an important role in determining restaurant popularity and success.

Future improvements could include incorporating additional attributes such as restaurant categories, pricing levels, and time-based trends to gain deeper insights into customer behavior.

10. Limitations & Future Work

Limitations

- Dataset limited to Yelp platform
- Some engagement metrics may not fully represent customer visits

Future Work

- Include restaurant category analysis
- Analyze trends over time
- Apply machine learning for success prediction